

Problem Setup

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Problem Setup

We are given the quadratic equation

$$x^2 + 4x + 4 = 0$$

and are asked to determine all real values of the variable x that satisfy this equation.

Strategy and Key Principles

The equation is a quadratic of the form $ax^2 + bx + c = 0$. Two standard methods apply:

1. **Factoring** – if the quadratic can be expressed as a perfect square or product of linear factors, set each factor to zero. 2. **Quadratic formula** – for any quadratic,
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Because the discriminant $b^2 - 4ac$ will turn out to be zero, the quadratic has a repeated (double) root, which can be found efficiently by factoring.

Detailed Solution

$$\begin{aligned}x^2 + 4x + 4 &= 0 \\(x + 2)^2 &= 0 \\x + 2 &= 0 \\x &= -2\end{aligned}$$

(Alternatively, applying the quadratic formula with $a = 1$, $b = 4$, $c = 4$:

$$\Delta = b^2 - 4ac = 16 - 16 = 0, \quad x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-4 \pm 0}{2} = -2.$$

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Final Answer

$$\boxed{-2}$$

Key Insights

- A quadratic with discriminant zero has a single (double) real root.
- Recognizing a perfect-square trinomial $(x+2)^2$ allows immediate factoring without the quadratic formula.
- A common mistake is to overlook the sign when completing the square; the correct factor is $(x+2)$, not $(x-2)$.